

Pin Kwang (PK) Tan

PH.D. CANDIDATE, UT AUSTIN NEUROSCIENCE

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Summary

Developing next-generation closed-loop visual neuroprosthetics in behaving macaques. Engineered a causal V1 "read-write" system (optogenetic stimulation + widefield imaging) enabling simultaneous perturbation and population-level readout. Demonstrated feature-specific perceptual effects (i.e., orientation) from V1 single-column stimulation. Building an image-computable encoding/decoding framework and evaluation harness toward prosthetic vision deployment. Modular architecture transferable to electrical microstimulation.

Python · MATLAB · optogenetics · widefield imaging · psychophysics · cortical viral delivery · primate surgery

Education

University of Texas at Austin

PH.D., NEUROSCIENCE (GPA: 3.99)

Austin, TX

Jan 2021 - Dec 2026

National University of Singapore

B.S. (HONS) LIFE SCIENCES (*summa cum laude*)

Singapore, SG

Aug 2014 - Aug 2018

Work Experience

UT Austin

PH.D. CANDIDATE (PI: DR. EYAL SEIDEMANN)

Austin, TX

Jan 2021 - Dec 2026

- Engineered a **bidirectional all-optical "read-write" system in V1** (optogenetic stimulation + widefield imaging) enabling simultaneous perturbation and population-level readout in awake, behaving macaques.
- Demonstrated feature-specific perceptual effects (e.g., orientation) driven by causal single-column V1 stimulation in behaving macaques; replicated across two animals.
- Designed and analyzed primate psychophysics experiments; quantified perceptual changes under stimulation vs control conditions with rigorous controls and model-based fits.
- Quantified stimulation-evoked network recruitment beyond the stimulated site (similarly tuned columns) and related physiological signatures to behavior.
- Built an image-computable encoding/decoding framework in Python (stimulus → V1 population activity → behavior), with baselines, evaluation harness, and modular architecture designed for closed-loop prosthetic deployment.
- Led iterative innovations in interface development, experimental execution, and analysis; mentored trainees on data QC and analysis workflows.

The N.1 Institute of Health

RESEARCH ASSISTANT (PI: DRS. CAMILO LIBEDINSKY, SHIH-CHENG YEN)

Singapore, SG

Jan 2019 - Dec 2020

- Designed and executed quantitative analysis of large-scale multi-electrode array recordings (Microprobes FMA, >300 channels; Plexon OmniPlex, 40 kHz) in primate dorsolateral prefrontal cortex (dlPFC) during a visual working memory task.
- Processed electrophysiological datasets, performing spike detection/sorting and precise alignment with behavioral/eye-tracking data (EyeLink)
- Developed and utilized MATLAB analysis pipelines for quantifying neural response properties to derive functional parcellation of the dlPFC (e.g., selectivity during visual/working memory periods, response latency)
- Applied statistical analyses (ANOVA, robust linear regression) and dimensionality reduction (PCA) to identify and validate **functional parcellations along the anterior-posterior axis**

Singapore Institute for Neurotechnology (SINAPSE)

RESEARCH INTERN (PI: DR. CAMILO LIBEDINSKY)

Singapore, SG

May 2019 - Aug 2019

- Analyzed stability and reliability of decoding algorithms for primate M1 signals driving a **BMI-controlled motorized wheelchair**; quantified neural code stability and decoder robustness across sessions — directly relevant to closed-loop reliability requirements.

Publications

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|------|---|---------------------|
| 2026 | Fast neural population dynamics in primate V1 captured by a genetically encoded voltage indicator
J. Zhou, Y. Chen, M. Whitmire, P. K. Tan , J. Wu, A. V., J. Robinson, W. Geisler, V. Pieribone, E. Seidemann | <i>Under review</i> |
| 2023 | Distinct lateral prefrontal regions are organized in an anterior-posterior functional gradient
P. K. Tan , C. Tang, R. Herikstad, A. Pillay, C. Libedinsky | <i>J. Neurosci.</i> |
| 2019 | Distinct genetic signatures of cortical and subcortical regions associated with human memory
P. K. Tan , E. Ananyev, P. J. Hsieh | <i>eNeuro</i> |

Honors & Awards

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|------|---|---------------|
| 2023 | Institute of Neuroscience Professional Development Award | Austin, TX |
| 2023 | Institute of Neuroscience Outstanding Poster Award | Austin, TX |
| 2017 | Agency for Science, Technology and Research, Singapore Chairman's Honors List | Singapore, SG |
| 2015 | Agency for Science, Technology and Research, Singapore (A*STAR) A*STAR Undergraduate Scholarship | Singapore, SG |

Professional activities

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| 2020 | Mentorship: Kate Pearce (MIT Junior), Charmaine Ter Li Min (NUS Senior) | Singapore, SG |
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